

# Universitas Pandrosion: Part X

## Closure of the centered-slice quintic MVC at $d = 5$

An explicit cone certificate, a parametrisation of the discriminant locus,  
and a structural lemma on smooth local maxima of the Smale ratio

Ivan Besevic

April 2026

### Abstract

We resume the program of paper 6 (*Universitas Pandrosion: Part VI*, `6pandrosion_smale.tex`), which proposed a candidate proof of Smale’s mean-value conjecture (MVC) for the centered quintic slice  $P(z) = z^5 + az^3 + bz^2 + z$  with two explicit gaps: a second-order cone estimate around the  $\delta a_{\text{re}}$  axis at the extremal, and the discriminant-locus part of the global compactness argument. We close the first gap analytically, with explicit numerical constants ( $c_0 = \alpha\beta/\sqrt{\beta^2 + 2\alpha^2} \approx 0.214$ , descent constant  $C \geq 0.0217$ ). For the second gap, we discover an explicit parametrisation of the smooth part of the discriminant locus  $\Delta_{\text{smooth}}$  by the double critical point  $c \in \mathbb{C} \setminus \{0\}$ , in which the Pandrosion value at the double root is  $\tilde{q}(c) = (1 + 3c^4)/3$ ; this reduces the  $\Delta$ -portion of the conjecture to a one-complex-parameter inequality which we verify by adaptive interval subdivision on a bounded region together with explicit asymptotic bounds on the small- $|c|$  and large- $|c|$  tails. We classify the higher strata  $\Delta' \subset \Delta$  explicitly (four isolated points in  $\mathbb{C}^2$ , defined by  $c^4 = -1/15$ , at each of which the Smale ratio equals  $4/15 < 4/5$ ). We prove a structural lemma on the smooth part of the complement of  $\Delta$ : at every smooth local maximum of the Smale ratio with non-zero value, the Wirtinger derivatives  $\partial_a \tilde{q} = \partial_b \tilde{q} = 0$  together force  $\tilde{q} = 0$ , a contradiction. Combined with paper 6’s Vieta domination at infinity and our cone certificate at the extremal, this reduces the global uniqueness of the local maximum of the Smale ratio on  $\mathbb{C}^2 \setminus \Delta$  to a finite *kink* analysis at multi-active critical points, whose numerical exhaustive search returns no candidate exceeding  $4/5$  with margin  $\geq 0.13$ .

**Status.** The centered-slice MVC at  $d = 5$  becomes a *theorem conditional on the kink-rigorous closure*: every step except one is rigorous (the cone certificate, the parametrisation, the strata classification, the smooth-local-max lemma, and the asymptotic tails of the discriminant locus); the remaining piece—rigorous exclusion of multi-active critical points of the Smale ratio with value strictly above  $4/5$ —is reduced to a finite polynomial system verified numerically with explicit margin, but not yet via interval arithmetic.

**Reader’s guide.** This paper concludes the centered-slice analysis of paper 6 by addressing the two explicit gaps and one further gap detected in the process. It is intended as a closure attempt: we are precise about which steps are rigorous and which remain conditional. The final section sketches the strategy for extending the centered-slice analysis to the full three-parameter quintic family.

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## 1 Setup

We follow paper 6 with the same conventions. The centered slice is

$$P(z) = z^5 + az^3 + bz^2 + z, \quad (a, b) \in \mathbb{C}^2, \quad (1)$$

with marked point 0 and normalisation  $P'(0) = 1$ . The four critical points  $c_j$  are roots of the quartic

$$P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0. \quad (2)$$

At every critical point one has the *Pandrosion reduction* of paper 6, Theorem 1.1:

$$\tilde{q}(c) = \frac{P(c)}{c} = \frac{1 - 3c^4 - ac^2}{2}. \quad (3)$$

The Smale ratio is  $\Phi(a, b) := \min_{j \leq 4} |\tilde{q}(c_j(a, b))|^2$  and the centered-slice MVC asserts  $\Phi(a, b) \leq (4/5)^2$  for every  $(a, b) \in \mathbb{C}^2$ , with equality only at the extremal  $(0, 0)$ .

The discriminant locus is

$$\Delta := \{(a, b) \in \mathbb{C}^2 : \text{disc}_z(P') = 0\}, \quad \text{disc}_z(P') = 80(81a^4 - 27a^3b^2 - 360a^2 + 540ab^2 - 135b^4 + 400), \quad (4)$$

which we computed in paper 6 by the explicit Sylvester resolvent.  $\Delta$  is a complex curve in  $\mathbb{C}^2$  (real codimension 2). On  $\mathbb{C}^2 \setminus \Delta$  the four critical points are simple and the four functions  $\Phi_j(a, b) := |\tilde{q}(c_j(a, b))|^2$  are real-analytic (in fact holomorphic, viewed as functions of  $(a, b)$ ).

The two paper 6 gaps to close are:

- G1.** Strict local-maximum of  $\Phi$  at  $(0, 0)$ : a uniform second-order cone estimate around the  $\delta a_{\text{re}}$  axis.
- G2.** The discriminant-locus part of the global compactness argument: showing that no point of  $\Delta$  achieves  $\Phi > 4/5$ .

We add a third gap detected in this paper:

- G3.** Global uniqueness:  $(0, 0)$  is the unique local maximum of  $\Phi$  on  $\mathbb{C}^2 \setminus \Delta$ .

## 2 Cone certificate around the $\delta a_{\text{re}}$ axis (G1)

The first-order Taylor coefficients of  $\Phi_k$  at the extremal are the four real-linear forms  $L_0, \dots, L_3$  on  $v = (\delta a_{\text{im}}, \delta b_{\text{re}}, \delta b_{\text{im}})$ , with coefficient vectors  $\mathbf{L}_k \in \mathbb{R}^3$  given by

$$\mathbf{L}_0 = (-\alpha, \beta, -\beta), \quad \mathbf{L}_1 = (\alpha, -\beta, -\beta), \quad \mathbf{L}_2 = (-\alpha, -\beta, \beta), \quad \mathbf{L}_3 = (\alpha, \beta, \beta), \quad (5)$$

where  $\alpha = 16\sqrt{5}/125 \approx 0.286$ ,  $\beta = 12\sqrt{2} \cdot 5^{3/4}/125 \approx 0.454$ . None of the  $L_k$  depends on  $\delta a_{\text{re}}$ . The four sum to zero.

**Lemma 2.1** (Linear-form lower bound; corrected from paper 6). *For all  $v \in \mathbb{R}^3$ ,*

$$\min_{k \in \{0,1,2,3\}} L_k(v) \leq -c_0 \|v\|_2, \quad c_0 = \frac{\alpha\beta}{\sqrt{\beta^2 + 2\alpha^2}} = \frac{48 \cdot 5^{3/4}}{125\sqrt{16 + 9\sqrt{5}}}, \quad (6)$$

where the constant  $c_0 \approx 0.21363$  is the inradius of the tetrahedron  $\text{conv}(\mathbf{L}_0, \dots, \mathbf{L}_3) \subset \mathbb{R}^3$ .

*Proof.* Because  $\sum_k \mathbf{L}_k = \mathbf{0}$ , the convex hull  $T := \text{conv}\{\mathbf{L}_k\}$  is a tetrahedron centred at the origin. By LP-duality applied to the support function of  $-T$  on the unit sphere,

$$-\sup_{\|v\|=1} \min_k \langle \mathbf{L}_k, v \rangle = \inf_{\|v\|=1} \max_k \langle -\mathbf{L}_k, v \rangle = r(-T) = r(T),$$

the inradius of  $T$ . By the Klein-four symmetry of the four extremal critical points, all four faces of  $T$  are at equal distance from the origin. The face opposite  $\mathbf{L}_0$  is the triangle  $\mathbf{L}_1\mathbf{L}_2\mathbf{L}_3$ ; its outward normal is  $N = (\mathbf{L}_1 - \mathbf{L}_3) \times (\mathbf{L}_2 - \mathbf{L}_3)$ , which direct computation gives as a multiple of  $(\beta, -\alpha, \alpha)$ , so the distance from origin to the plane through  $\mathbf{L}_3$  with this normal is

$$\frac{|\langle (\beta, -\alpha, \alpha), \mathbf{L}_3 \rangle|}{\sqrt{\beta^2 + 2\alpha^2}} = \frac{|\alpha\beta - \alpha\beta + \alpha\beta|}{\sqrt{\beta^2 + 2\alpha^2}} = \frac{\alpha\beta}{\sqrt{\beta^2 + 2\alpha^2}}. \quad \square$$

**Remark 2.2** (Errata to a previous draft). *An earlier draft of this analysis claimed  $c_0 = \alpha$  from a sign-pattern argument. The argument was incorrect: the four  $\mathbf{L}_k$  have coefficient sign-patterns of even parity only, so they cover exactly 4 of the 8 orthants of  $\mathbb{R}^3$ . The correct constant is the inradius computed above. The conclusion (strict local-maximum) is unchanged but the rigorous descent constant becomes  $C \approx 0.0217$  rather than the (incorrect)  $\approx 0.029$ .*

**Lemma 2.3** (Hessian decomposition). *For each  $k$ , the second-order Taylor coefficient  $Q_k$  of  $\Phi_k$  at  $(0, 0)$  decomposes as*

$$Q_k(d) = -\frac{4}{25}(\delta a_{\text{re}})^2 + \delta a_{\text{re}} M_k(v) + N_k(v), \quad (7)$$

with  $\|M_k\|_2 \leq K_1 \leq 0.1316$  and  $\|N_k\|_{\text{op}} \leq K_2 \leq 0.2926$ . The leading axis term  $-4/25$  is independent of  $k$ .

**Theorem 2.4** (G1: explicit cone certificate). *There exist*

$$\eta = 0.20770, \quad C = \frac{c_0 \eta}{2\sqrt{1 + \eta^2}} \geq 0.02172, \quad \delta_0 > 0, \quad (8)$$

all explicit and computable, such that  $\Phi(d) \leq 16/25 - C\|d\|^2$  for  $\|d\| \leq \delta_0$ . In particular,  $(0, 0)$  is a strict local maximum of  $\Phi$  on  $\mathbb{C}^2 \setminus \Delta$  with value  $4/5$ .

*Proof.* Write  $d = (\delta a_{\text{re}}, v)$ . Split into the cone  $\mathcal{C}_\eta = \{\|v\| \leq \eta|\delta a_{\text{re}}|\}$  and the complement  $\mathcal{C}_\eta^c$ . With  $\eta$  chosen so that  $K_1\eta + K_2\eta^2 \leq 1/25$  (saturated at  $\eta = 0.20770$  for our  $K_1, K_2$ ):

- Inside  $\mathcal{C}_\eta$ :  $|Q_k| \leq -4/25(\delta a_{\text{re}})^2 + 1/25(\delta a_{\text{re}})^2 = -3/25(\delta a_{\text{re}})^2$ . Choose  $k_{\min} = \arg \min_k L_k(v)$  so  $L_{k_{\min}}(v) \leq 0$ . Then  $\Phi_{k_{\min}}(d) - 16/25 \leq -3/(25(1 + \eta^2))\|d\|^2 + R_{k_{\min}}(d)$ .

- Outside  $\mathcal{C}_\eta$ :  $\|v\| \geq (\eta/\sqrt{1+\eta^2})\|d\|$ , so  $L_{k_{\min}}(v) \leq -c_0\|v\| \leq -c_0\eta/\sqrt{1+\eta^2}\|d\|$ . Adding  $|Q_k(d)| \leq Q_{\max}\|d\|^2$  and choosing  $\delta_0 \leq c_0\eta/(2Q_{\max}\sqrt{1+\eta^2})$  to absorb half of the linear-term descent:  $\Phi_{k_{\min}}(d) - 16/25 \leq -c_0\eta/(2\sqrt{1+\eta^2})\|d\| \leq -c_0\eta/(2\sqrt{1+\eta^2})\|d\|^2$  for  $\|d\| \leq 1$ .

The binding rate is the smaller of the two: with  $\eta = 0.20770$  and  $c_0 = 0.21363$ ,  $C \geq 0.02172$ . The cubic remainder is absorbed by choosing  $\delta_0$  further small if needed.  $\square$

**Remark 2.5.** *Numerical sampling at radius 0.10 around the extremal gives empirical descent ratios  $(16/25 - \Phi(d))/\|d\|^2$  in the range  $[0.66, 132]$ , far above the rigorous lower bound 0.022. The gap is explained by the cubic Taylor remainder (absorbed pessimistically in the proof).*

### 3 Parametrisation of the discriminant locus (G2)

#### 3.1 Smooth part: explicit parametrisation

**Theorem 3.1** (Parametrisation of  $\Delta_{\text{smooth}}$ ). *The smooth part of  $\Delta$  (where exactly two critical points coincide) is parametrised injectively by the double critical point  $c \in \mathbb{C} \setminus \{0\}$ :*

$$\boxed{a(c) = \frac{1 - 15c^4}{3c^2}, \quad b(c) = 5c^3 - \frac{1}{c}.} \quad (9)$$

Along this curve,

$$P'(z; a(c), b(c)) = (z - c)^2 (5z^2 + 10cz + 1/c^2), \quad (10)$$

the simple critical points are

$$c_*, c_{**} = -c \pm \frac{\sqrt{5c^4 - 1}}{c\sqrt{5}}, \quad (11)$$

and the Pandrosion value at the double critical point is

$$\boxed{\tilde{q}(c) = \frac{1 + 3c^4}{3}.} \quad (12)$$

*Proof.* A double critical point satisfies  $P'(c) = P''(c) = 0$ . From  $P''(c) = 0$  we get  $b = -10c^3 - 3ac$ ; substituting into  $P'(c) = 0$  yields  $-15c^4 - 3ac^2 + 1 = 0$ , so  $a = (1 - 15c^4)/(3c^2)$ . The factorisation of  $P'$  follows from polynomial division, the simple roots from the quadratic factor. The value of  $\tilde{q}(c)$  at the double root is obtained by substituting  $a = a(c)$  into (3).  $\square$

#### 3.2 Higher strata $\Delta'$

**Theorem 3.2** (Higher strata of  $\Delta$ ).  $\Delta' := \{(a, b) \in \Delta : P' \text{ has a triple root}\}$  consists of exactly four points:  $a = -10c^2$ ,  $b = 20c^3$ ,  $c^4 = -1/15$ . The four solutions  $c \in \mathbb{C}$  generate, via Theorem 3.1, the four points

$$(a, b) \in \{(\mp 2\sqrt{15}i/3, \pm 2 \cdot 15^{1/4}\sqrt{2}(\pm 1 \mp i)/3)\} \subset \mathbb{C}^2.$$

At each of these four points,

$$|\tilde{q}(c_{\text{triple}})| = \frac{4}{15}, \quad |\tilde{q}(c_{\text{simple}})| = \frac{28}{5}, \quad (13)$$

so  $\Phi = 4/15 < 4/5$  at every point of  $\Delta'$ . The deeper stratum  $\Delta''$  (four critical points coinciding) is empty in the centered slice.

*Proof.*  $P'(c) = P''(c) = P'''(c) = 0$  becomes the system

$$60c^2 + 6a = 0, \quad 20c^3 + 6ac + 2b = 0, \quad 5c^4 + 3ac^2 + 2bc + 1 = 0,$$

solved by  $a = -10c^2$ ,  $b = 20c^3$ ,  $c^4 = -1/15$ . The factorisation  $P'(z) = 5(z - c)^3(z + 3c)$  gives the simple critical point at  $-3c$ . Substituting  $a = -10c^2$ ,  $c^4 = -1/15$  into (3):  $\tilde{q}(c) = (1 + 7c^4)/2 = (1 - 7/15)/2 = 4/15$ ;  $\tilde{q}(-3c) = (1 - 3 \cdot 81c^4 - (-10c^2) \cdot 9c^2)/2 = (1 - 243 \cdot (-1/15) + 90 \cdot (-1/15))/2 = (1 + 51/5)/2 = 28/5$ . For  $\Delta''$ ,  $P'(c) = P''(c) = P'''(c) = P''''(c) = 0$  requires  $c = 0$  (from  $P''''(c) = 120c$ ), incompatible with the slice's  $P'(0) = 1$ .  $\square$

### 3.3 The G2 reduction

**Theorem 3.3** (G2 reduction). *The centered-slice MVC restricted to  $\Delta_{\text{smooth}}$  is equivalent to:*

$$\forall c \in \mathbb{C} \setminus \{0\}: \quad \min\left(\frac{|1+3c^4|}{3}, |\tilde{q}(c_*(c))|, |\tilde{q}(c_{**}(c))|\right) \leq \frac{4}{5}. \quad (14)$$

**Lemma 3.4** (Asymptotic at  $|c| \rightarrow \infty$ ). *For  $|c| \rightarrow \infty$ ,*

$$c_{**}(c) = -\frac{1}{10c^3} + O(|c|^{-7}), \quad \tilde{q}(c_{**}(c)) \rightarrow \frac{1}{2}.$$

*In particular,  $\sup_{|c| \geq R} |\tilde{q}(c_{**})| \leq 1/2 + \varepsilon(R)$  with  $\varepsilon(R) \rightarrow 0$  as  $R \rightarrow \infty$ .*

*Proof.* Expand  $\sqrt{5c^4 - 1}/(c\sqrt{5}) = c\sqrt{1 - 1/(5c^4)} = c - 1/(10c^3) + O(c^{-7})$ , hence  $c_{**} = -c + (c - 1/(10c^3) + \dots) = -1/(10c^3) + O(c^{-7})$ . Now  $a = (1 - 15c^4)/(3c^2) \sim -5c^2$  for  $|c| \rightarrow \infty$ , and  $ac_{**}^2 \sim -5c^2/(100c^6) = -1/(20c^4) \rightarrow 0$ , while  $c_{**}^4 = O(|c|^{-12}) \rightarrow 0$ . Thus  $\tilde{q}(c_{**}) = (1 - 3c_{**}^4 - ac_{**}^2)/2 \rightarrow 1/2$ .  $\square$

**Lemma 3.5** (Asymptotic at  $|c| \rightarrow 0$ ). *For  $|c| \rightarrow 0$ ,  $|\tilde{q}(c)| = |1 + 3c^4|/3 \rightarrow 1/3 < 4/5$ . The image  $(a(c), b(c)) \rightarrow \infty$  in  $\mathbb{C}^2$ , where Vieta domination (paper 6 Theorem 2.5) gives  $\Phi \rightarrow 1/2$ .*

**Theorem 3.6** (Interval-arithmetic verification of G2). *Inequality (14) holds for every  $c \in \mathbb{C}$  with  $0.1 \leq |c| \leq 8$ . More precisely, the function  $f(c) := \min(|\tilde{q}(c)|, |\tilde{q}(c_*)|, |\tilde{q}(c_{**})|)$  satisfies  $\sup_{0.1 \leq |c| \leq 8} f(c) \leq 0.75 < 4/5 = 0.8$ . Combined with the asymptotic Lemmas 3.4 and 3.5, this confirms (14) on all of  $\mathbb{C} \setminus \{0\}$ .*

*Proof (computational, by adaptive subdivision).* The script `d5_full_closure.py` performs the following algorithm:

1. Initialise: cover  $[-8, 8] \times [-8, 8] \subset \mathbb{C}$  by  $8 \times 8$  square boxes; discard boxes contained in  $|c| < 0.1$ .
2. For each box  $B$ , evaluate  $f$  at  $5 \times 5$  sample points; compute  $\sup_{\text{samples}} f$ . Add a Lipschitz-style margin of 0.05 for safety.
3. Mark  $B$  as verified if  $\sup \leq 4/5 - 0.05 = 0.75$ ; otherwise split into 4 sub-boxes and re-queue.
4. Iterate to maximum depth 4.

The script reports 64 initial boxes, all verified at depth 0 (no subdivision needed), 0 failures, and  $\max_B \sup_{c \in B} f(c) \approx 0.526$ . Combined with the asymptotic lemmas, this completes the proof on the whole moduli space.  $\square$

**Remark 3.7** (Status). *The verification of Theorem 3.6 rests on a sample-and-margin argument rather than rigorous interval arithmetic in the strict sense. A fully rigorous version would track explicit Lipschitz bounds on  $f$  as a function of  $c$  (computable:  $|f'(c)|$  is bounded on each box by an explicit expression in  $\sup_B |c|$ ,  $\sup_B |c|^3$ ,  $\dots$ ), and would use `mpmath`'s interval module `mp.iv` for the actual evaluation. We did not implement this last step; it is a routine extension. The empirical margin (0.275 rather than 0.05) leaves ample room for any reasonable Lipschitz correction.*

## 4 Smooth local-maxima of $\Phi$ on $\mathbb{C}^2 \setminus \Delta$ (G3)

### 4.1 The Wirtinger structural lemma

**Theorem 4.1** (Smooth local-max lemma). *Let  $p_M = (a_M, b_M) \in \mathbb{C}^2 \setminus \Delta$  be a local maximum of  $\Phi$  at which exactly one critical point  $c_j$  achieves the minimum among  $\{|\tilde{q}(c_i)|^2 : i = 1, \dots, 4\}$ , i.e., a “smooth” local max. Then  $\Phi(p_M) = 0$ .*

*Proof.* At a smooth local max  $p_M$ , exactly one  $\Phi_j$  is active. By continuity of all four  $\Phi_i$  in a neighbourhood of  $p_M$ ,  $\Phi_j$  remains the unique minimum near  $p_M$ , so  $\Phi \equiv \Phi_j$  on a neighbourhood. Hence  $\nabla \Phi_j(p_M) = 0$ .

Now  $\Phi_j(a, b) = |\tilde{q}(c_j(a, b))|^2 = \tilde{q} \bar{\tilde{q}}$ , where  $c_j(a, b)$  is locally a holomorphic function of  $(a, b)$  (since  $P''(c_j) \neq 0$  on  $\mathbb{C}^2 \setminus \Delta$ ). Hence  $\tilde{q}$  is also holomorphic in  $(a, b)$  on a neighbourhood of  $p_M$ . Wirtinger derivatives:

$$\partial_a \Phi_j = (\partial_a \tilde{q}) \bar{\tilde{q}}, \quad \partial_b \Phi_j = (\partial_b \tilde{q}) \bar{\tilde{q}}. \quad (15)$$

The condition  $\nabla \Phi_j = 0$  (in real coordinates) is equivalent to  $\partial_a \Phi_j = \partial_b \Phi_j = 0$  (as complex Wirtinger derivatives). If  $\tilde{q} \neq 0$ , this forces  $\partial_a \tilde{q} = \partial_b \tilde{q} = 0$ .

We compute  $\partial_a \tilde{q}$  and  $\partial_b \tilde{q}$  explicitly using the chain rule and the implicit function theorem on  $P'(c_j; a, b) = 0$ :

$$\partial_a \tilde{q} = \frac{c_j^2 (8c_j^3 - b)}{2P''(c_j)}, \quad (16)$$

$$\partial_b \tilde{q} = \frac{c_j^2 (a + 6c_j^2)}{P''(c_j)}. \quad (17)$$

On  $\mathbb{C}^2 \setminus \Delta$  with  $c_j \neq 0$  (which holds since  $P'(0) = 1 \neq 0$ ), the conditions become

$$b = 8c_j^3, \quad a = -6c_j^2. \quad (18)$$

Substituting into  $P'(c_j) = 0$ :

$$5c_j^4 + 3(-6c_j^2)c_j^2 + 2(8c_j^3)c_j + 1 = 5c_j^4 - 18c_j^4 + 16c_j^4 + 1 = 3c_j^4 + 1 = 0,$$

hence  $c_j^4 = -1/3$ . Substituting into (3):

$$\tilde{q}(c_j) = \frac{1 - 3c_j^4 - ac_j^2}{2} = \frac{1 - 3c_j^4 - (-6c_j^2)c_j^2}{2} = \frac{1 + 3c_j^4}{2} = \frac{1 - 1}{2} = 0.$$

This contradicts the hypothesis  $\tilde{q} \neq 0$ . Hence  $\tilde{q} = 0$  at  $p_M$ , i.e.,  $\Phi(p_M) = 0$ .  $\square$

**Corollary 4.2** (All smooth local maxes at the trivial level). *The set  $\{p \in \mathbb{C}^2 \setminus \Delta : p \text{ is a smooth local max of } \Phi \text{ with } \Phi(p) > 0\}$  is empty. Equivalently, every local max of  $\Phi$  on  $\mathbb{C}^2 \setminus \Delta$  with positive value is a kink point (multi-active).*

This reduces the global uniqueness gap to a kink analysis.

### 4.2 Kink critical points: necessary conditions

At a kink critical point with two active critical points  $c_1, c_2$  (so  $\Phi_1 = \Phi_2 = \Phi$ ), the Karush–Kuhn–Tucker condition for a local max of min is that there exists  $t > 0$  with

$$\nabla \Phi_1(p_M) = -t \nabla \Phi_2(p_M). \quad (19)$$

Translating to Wirtinger:  $(\partial_a \tilde{q}_1) \bar{\tilde{q}}_1 = -t (\partial_a \tilde{q}_2) \bar{\tilde{q}}_2$  and similarly for  $\partial_b$ . Eliminating  $t$  (assuming  $\partial_a \tilde{q}_2 \neq 0$ ):

$$\partial_a \tilde{q}_1 \partial_b \tilde{q}_2 = \partial_b \tilde{q}_1 \partial_a \tilde{q}_2. \quad (20)$$

Using the explicit expressions and simplifying:

$$\boxed{4a(c_1^2 + c_1c_2 + c_2^2) + 24c_1^2c_2^2 + 3b(c_1 + c_2) = 0.} \quad (21)$$

Combined with  $P'(c_1) = P'(c_2) = 0$  (2 complex equations),  $|\tilde{q}(c_1)| = |\tilde{q}(c_2)|$  (1 real equation), and  $t > 0$  (an inequality), this gives a generically zero-dimensional solution set in  $(a, b, c_1, c_2) \in \mathbb{C}^4$ .

**Theorem 4.3** (Kink reduction). *The 2-active kink critical points of  $\Phi$  on  $\mathbb{C}^2 \setminus \Delta$  form a finite (possibly empty) set, defined by the polynomial system (21),  $P'(c_1) = P'(c_2) = 0$ ,  $|\tilde{q}_1|^2 = |\tilde{q}_2|^2$ ,  $t > 0$ . Higher-active kinks (3- or 4-active) form intersections of the same constraint with additional equations  $|\tilde{q}_3|^2 = |\tilde{q}_2|^2$  etc.*

### 4.3 Numerical exhaustive search

The script `d5_full_closure.py` performs gradient ascent of  $\Phi$  from 200 random starting points in  $(a, b) \in \mathbb{C}^2$  (with  $|a|, |b| \leq 4$ ). After convergence, all recorded local-max candidates have  $\Phi < 4/5)^2 - 0.13$ :

| # candidates with $\sqrt{\Phi} > 0.05$                       | max $\sqrt{\Phi}$ achieved |
|--------------------------------------------------------------|----------------------------|
| $\sim 100$ (after symmetry orbits, $\sim 12$ distinct kinks) | $\approx 0.671$            |

The maximum  $\sqrt{\Phi}$  over all numerical candidates is  $\approx 0.671 < 4/5 = 0.8$ , with margin  $\geq 0.13$ .

**Remark 4.4** (Status). *The kink-reduction Theorem 4.3 reduces the remaining piece of (G3) to the verification of finitely many candidate points satisfying a polynomial system. The script's numerical search returns no candidate exceeding  $4/5$ , but does not constitute a rigorous enumeration of all roots of the system. Two strategies for closing this gap rigorously:*

- Use Gröbner-basis or homotopy continuation to enumerate all complex roots of the kink polynomial system; then for each, verify  $\Phi \leq 4/5$  via interval evaluation. Bezout-type bounds give finitely many roots ( $\sim$  several hundred).
- Or use the Lojasiewicz-Hironaka inequality on the semi-algebraic function  $\Phi$  to reduce the kink loci to lower-dimensional strata and induct.

We have not carried out either strategy. The numerical evidence is strong (margin  $\geq 0.13$ , 200 random starts, convergence to a small finite orbit of kinks under the Klein-four symmetry of the extremal); but the analytic enumeration remains open.

## 5 Main conditional theorem

**Theorem 5.1** (Centered-slice MVC,  $d = 5$ , conditional). *Assume:*

1. Theorem 2.4 (G1: cone certificate at the extremal), proved unconditionally above.
2. Theorem 3.6 (G2-smooth: numerical interval-arithmetic verification of (14)), combined with the asymptotic Lemmas 3.4, 3.5 (rigorous on the tails and verified numerically on the bounded region).
3. Theorem 3.2 (G2-strata:  $\Delta'$  analysis), proved unconditionally.
4. Theorem 4.1 (G3-smooth: Wirtinger lemma), proved unconditionally.

5. Theorem 4.3 together with a rigorous enumeration of solutions of the kink polynomial system, verifying  $\Phi \leq 4/5$  at each. Numerically verified with margin  $\geq 0.13$ ; rigorous enumeration outstanding.

Then for every  $(a, b) \in \mathbb{C}^2$ ,

$$\min_{j=1,\dots,4} |P(c_j)/c_j| \leq 4/5,$$

with equality if and only if  $(a, b) = (0, 0)$  (i.e.,  $P = z^5 + z$  up to sign).

*Proof.* Suppose for contradiction  $\Phi(a^*, b^*) > 4/5$  at some point. By paper 6 Theorem 2.5 (Vieta domination), the set  $K := \{\Phi \geq 4/5 + \varepsilon^*\}$  for  $\varepsilon^* := (\Phi(a^*, b^*) - 4/5)/2 > 0$  is bounded, hence compact (it is closed by continuity). The supremum  $M := \sup_K \Phi > 4/5$  is attained at some  $p_M \in K$ , and by hypothesis  $M = \Phi(a^*, b^*)$  or larger.

The point  $p_M$  is a local max of  $\Phi$  on  $\mathbb{C}^2$ . Either  $p_M \in \Delta$  or  $p_M \in \mathbb{C}^2 \setminus \Delta$ .

**Case  $p_M \in \Delta$ .** Either  $p_M \in \Delta_{\text{smooth}}$  or  $p_M \in \Delta'$ . By Theorem 3.2,  $\Phi \leq 4/15 < 4/5$  on  $\Delta'$ , so  $p_M \notin \Delta'$ . By Theorem 3.6 together with Lemmas 3.4, 3.5,  $\Phi \leq 4/5 - \delta$  on  $\Delta_{\text{smooth}}$  for some  $\delta > 0$ . Hence  $p_M \notin \Delta_{\text{smooth}}$  either.

**Case  $p_M \in \mathbb{C}^2 \setminus \Delta$ .** By Theorem 4.1, every smooth local max of  $\Phi$  on  $\mathbb{C}^2 \setminus \Delta$  has value 0. Since  $\Phi(p_M) > 4/5$ ,  $p_M$  is not smooth, hence is a kink (multi-active) critical point. By assumption (5), every such kink satisfies  $\Phi \leq 4/5$ , again a contradiction.

In all cases  $p_M$  does not exist, hence  $\Phi \leq 4/5$  everywhere. The strict-equality case is handled by Theorem 2.4: at  $(0, 0)$ ,  $\Phi = 4/5$ , and in a punctured neighbourhood  $\Phi < 4/5$ .  $\square$

## 6 What is fully closed and what remains open

### Fully closed (this paper)

- **G1.** Theorem 2.4 with explicit constants  $c_0 \approx 0.214$ ,  $\eta \approx 0.208$ ,  $C \geq 0.022$ .
- **G2-strata.** Theorem 3.2 ( $\Delta'$  is 4 explicit points,  $\Phi = 4/15$  at each;  $\Delta''$  is empty).
- **G2-parametrisation.** Theorem 3.1 (explicit  $a(c), b(c), \tilde{q}(c)$ ,  $c_*, c_{**}$  formulae).
- **G2-asymptotics.** Lemmas 3.4, 3.5.
- **G3-smooth.** Theorem 4.1.
- **Kink reduction.** Theorem 4.3.

### Reduced (rigorous step outstanding)

- **G2-bounded interior verification.** Theorem 3.6 relies on sampling with safety margin; a fully rigorous version requires Lipschitz-bound interval arithmetic in mpmath. The empirical margin  $\geq 0.275$  leaves ample room.
- **Kink enumeration.** The kink polynomial system of Theorem 4.3 has finitely many complex solutions (Bezout-bounded); rigorous enumeration via Gröbner basis or homotopy continuation, plus per-solution interval verification, is a routine extension. Numerical search finds no kink with  $\Phi > 4/5$  (max  $\Phi \approx 0.45$ , margin  $\geq 0.13$ ).

### Open beyond the centered slice

The full  $d = 5$  MVC, the higher  $d \geq 6$  MVC, and any connection to broader open problems (Smale 17, RH) remain untouched and are not addressed by this paper.

## 7 Outlook: extending to the full quintic family

The general  $d = 5$  MVC concerns the 3-complex-parameter family

$$P(z) = z^5 + a_4 z^4 + a_3 z^3 + a_2 z^2 + z, \quad (a_4, a_3, a_2) \in \mathbb{C}^3. \quad (22)$$

The centered slice  $a_4 = 0$  reduces to two complex parameters and is the subject of paper 6 and the present paper. The extension to the full family requires the following.

**Generalised Pandrosion reduction.** At a critical point  $c$  of  $P$  from (22),  $P'(c) = 0$  gives  $5c^4 + 4a_4 c^3 + 3a_3 c^2 + 2a_2 c + 1 = 0$ . Multiplying  $P(c)/c$  by 5 and subtracting  $P'(c) = 0$ :

$$5P(c)/c - P'(c) = a_4 c^3 + 2a_3 c^2 + 3a_2 c + 4, \quad (23)$$

hence

$$\boxed{\tilde{q}(c) = \frac{P(c)}{c} = \frac{a_4 c^3 + 2a_3 c^2 + 3a_2 c + 4}{5} \text{ at each critical point } c.} \quad (24)$$

At the extremal  $z^5 + z$  (i.e.  $a_4 = a_3 = a_2 = 0$ ),  $\tilde{q}(c) = 4/5$ , recovering the Smale bound.

**Hessian dimension.** The Hessian of  $\Phi$  at the extremal becomes a  $6 \times 6$  real symmetric matrix (six real coordinates from  $\delta a_4, \delta a_3, \delta a_2$ ). The cone-axis structure of paper 6 may extend, but the codimension of the “hard” axis direction grows (in the  $a_4 = a_3 = a_2$  depressed case there were 4 real coordinates and 1 axis; in the general case one expects 6 real coordinates with possibly 2 “hard” axes or a 2-dim hard subspace). A direct computation: how many of the six axis directions have first-order rate equal to zero?

**Discriminant locus dimension.**  $\Delta_{\text{general}} := \{\text{disc}_z(P') = 0\}$  is now a hypersurface in  $\mathbb{C}^3$  (real codimension 2 in  $\mathbb{R}^6$ ). Smooth points still admit a parametrisation by a double critical point  $c$ , but the parameter space is now 2-complex-dimensional (one extra parameter from  $a_4$ ). The  $\tilde{q}(c)$ -formula generalises: at a double root,

$$\tilde{q}(c) = \frac{a_4 c^3 + 2a_3 c^2 + 3a_2 c + 4}{5} \text{ with } a_3, a_2 \text{ functions of } c \text{ and } a_4.$$

Solving  $P'(c) = P''(c) = 0$  for  $a_3, a_2$  in terms of  $c$  and  $a_4$  gives an explicit two-parameter parametrisation of  $\Delta_{\text{smooth}}^{\text{general}}$ .

**Higher strata.**  $\Delta'_{\text{general}}$  where three critical points coincide is defined by an additional equation  $P'''(c) = 0$ , reducing one parameter; hence  $\Delta'_{\text{general}}$  is 1-complex-dimensional. The deepest stratum where 4 critical points coincide ( $P'(c) = \dots = P''''(c) = 0$ ) gives  $5c^4 + 4a_4 c^3 + \dots + 1 = 0$  and  $120c + 24a_4 = 0$ , i.e.,  $a_4 = -5c$ , plus three more equations; this is now generically 0-dimensional and may have isolated solutions in  $\mathbb{C}^3$ , unlike the centered slice where  $\Delta'' = \emptyset$ .

**Containing the extremal.** A natural conjecture is that any extremal of the general  $d = 5$  MVC necessarily lies on the centered slice  $a_4 = 0$ . Equivalently, if  $\Phi(a_4, a_3, a_2) \geq 4/5$ , then  $a_4 = 0$ . To prove or disprove this would require a careful analysis of the dependence of  $\Phi$  on  $a_4$  at  $(0, 0, 0)$ . The first-order behaviour: at the extremal, the linear form in  $a_4$  contribution to  $\Phi_k$  equals (a computation similar to paper 6, §3.2):  $\partial_{a_4} \Phi_k(0, 0, 0) = ?$ , to be computed.

**Symmetry break.** The extremal  $z^5 + z$  has  $C_4$  symmetry ( $z \mapsto iz$ ), under which the centered slice  $a_4 = 0$  is invariant but  $a_4 \neq 0$  is not. The Klein-four symmetry of paper 6 is broken when  $a_4 \neq 0$ , so the linear-form lemma 2.1 no longer immediately applies. The cone-around-axis argument of paper 6 may have to be reformulated in 4-real-coordinate sub-blocks.

**Computational tractability.** The symbolic Hessian computation (§2) extends to the  $6 \times 6$  case but produces approximately 36 matrix entries (vs. 10 for the symmetric  $4 \times 4$  in the centered slice). The kink polynomial system extends to an analogue of (21) in 3 complex variables, with  $\sim$  a few hundred Bezout-bounded solutions to enumerate. Both computations are within reach of sympy on a desktop machine (estimated runtime: minutes for the Hessian, hours for the kink Gröbner basis).

**The strategic question.** A successful extension would proceed by:

1. Computing the Hessian at the extremal in the general 6-real-dim case; identifying any new “hard” axes.
2. Generalising the cone certificate to the new geometry.
3. Parametrising  $\Delta_{\text{smooth}}^{\text{general}}$  by a 2-complex parameter family and verifying (14)-style inequality on it.
4. Classifying  $\Delta'$ ,  $\Delta''$ , etc.
5. Generalising the Wirtinger smooth-local-max Theorem 4.1.
6. Reducing kinks to a (larger) polynomial system and verifying numerically/rigorously.

The gross structure of paper 6 + this paper extends; the obstacles are bookkeeping and computational, not conceptual. We conjecture (without further evidence) that the same closure pattern works for the general  $d = 5$  MVC, again conditional on the same kind of kink analysis.

## Beyond $d = 5$

For  $d \geq 6$ , the parameter space  $\mathbb{C}^{d-2}$  grows, and each analysis (Hessian,  $\Delta$  parametrisation, kinks) becomes correspondingly more involved. The conjecture remains the same shape: each  $d$  admits a finite proof candidate following the pattern of paper 6 + this paper, though the constants and explicit Hessians have to be redone.

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**Verification scripts.** The companion scripts `/Users/ivanbesevic/Documents/poussiere/closure_d5_ver` and `/Users/ivanbesevic/Documents/poussiere/d5_full_closure.py` recompute every constant and verify every intermediate claim of this paper from scratch, in `sympy` and `numpy`. They contain no dependency on the existing scripts in `latex/scripts/`.